

A.12 Jahn-Teller Distortions in LiNiO_2

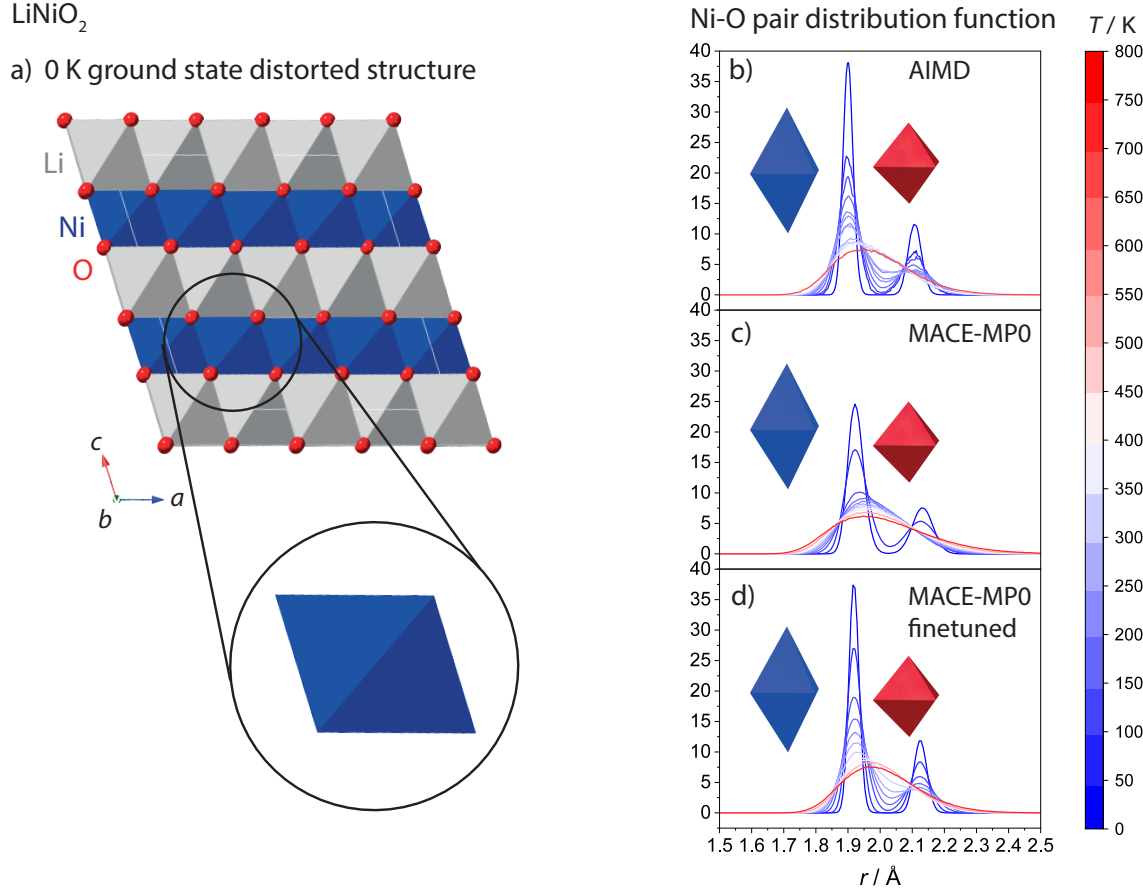


Figure S17: a) Ground state structure of LNO. Magnified insert shows a NiO_6 octahedron with Jahn-Teller distortion producing long and short axes. b) Ni–O pair distribution functions (PDF) as a function of ion separation computed by AIMD. (212) Line colour corresponds to trajectory temperature as shown in the colorbar. c) PDFs computed by MACE-MP-0b3. Inserts indicate shapes of NiO_6 octahedra (blue: distorted, red: undistorted). d) PDFs computed by fine-tuned MACE model as described in text. The same temperature values are used in panels c) and d), slightly fewer values are used in AIMD but spanning a similar range of temperature.

LiNiO_2 (LNO) is an important material in lithium-ion battery research, serving as a model for future high-performance cathode materials with reduced or no cobalt content. Structural and chemical degradation of these materials is a key challenge, so understanding their thermodynamic and dynamic properties is critical. In particular, the presence of ordered regions of Jahn-Teller distorted NiO_6 octahedra may influence electron and Li-ion transport. Recent *ab initio* molecular dynamics (AIMD) and variable-temperature X-ray diffraction (VT-XRD) measurements have suggested a phase transition from a low-temperature phase with Jahn-Teller distortions to a dynamic phase without permanent Jahn-Teller distortions at high temperature (212). Above this displacive phase transition the material does not exhibit Jahn-Teller distortions either in the time-averaged bulk structure or in instantaneous snapshots of the local structure. A similar displacive transition was also observed in the related sodium-ion battery cathode material NaNiO_2 . (213)

We have investigated the ability of MACE-MP-0 to reproduce the temperature dependence of the structural distortions in LNO. We used a 256-atom supercell, initialized from the established DFT ground state structure which has distorted NiO_6 octahedra with zigzag long-ranged ordering of their long axes (see fig. S17a). We performed NPT-ensemble dynamics using MACE-MP-0 without a D3 correction, in line with the earlier AIMD simulations at the $\text{PBE}+U$ level of theory.

Trajectories were propagated for approximately 250 ps at 13 temperature values from 25 K to 1000 K. These trajectories were used to analyse the phase transition behaviour (fig. S17c), and a sample of 104 configurations was used to fine-tune a new model following the protocol outlined in sec. C.2. Specifically, we extracted configurations from each trajectory at 20 ps intervals, after discarding the first 100 ps for equilibration. The performance of the fine-tuned model is compared with that of the original MACE-MP-0 in the next section.

Performance

The MACE-MP-0 model demonstrates stable dynamics during a heating trajectory from 0 K to 1000 K with temperature increments of 10 K/ps, time step 1 fs. Long simulations (at least 500 ps) could be run at 1000 K without significant energy drift or noticeably unphysical behavior.

The Ni-O pair distribution functions predicted by both AIMD and MACE-MP-0 (Fig. S17b and c) show two peaks at low temperature corresponding to the short and long Ni-O bonds in Jahn-Teller distorted octahedra. At higher temperatures one broad asymmetric peak appears, which is characteristic of octahedra in a displacive high-temperature phase. (212) MACE-MP-0 thus correctly predicts a phase transition from Jahn-Teller-distorted to undistorted octahedra on heating. However, it significantly underestimates the temperature at which the transition occurs. We diagnose the onset of the transition at the lowest temperature where the PDF has only one maximum and the second peak (at longer bond lengths) becomes a shoulder on the short-bond peak. Above this temperature the trajectories show dynamic Jahn-Teller distortions, i.e. reorientations of the long O-Ni-O axes over time. This onset occurs between 50 and 100 K in the MACE-MP-0 simulations, compared to ca. 250 K with AIMD. (212) The transition is complete when the Ni-O PDF exhibits only one broad asymmetric peak without a shoulder. MACE-MP-0 predicts that this completion temperature is below 150 K, compared with 350 K in AIMD.

By contrast, the fine-tuned model predicts an onset temperature between 250 and 300 K, and the transition to complete by 350 K, both in excellent agreement with the AIMD results. This improvement suggests that the deficiency of the original model results from undersampling of important configurations in the training data.

Similarity statement

Battery materials are well represented in the MP database. 1393 structures in the training set contain Li, Ni, or O atoms, 143 contain all three elements, and 23 have the exact formula LiNiO_2 representing different crystal structures. However, many of these structures are obtained from geometry optimization and hence neglect the temperature dependence of the equilibrium geometry. The substantial improvement of MACE-MP-0 performance for this system after fine-tuning on variable-temperature trajectories suggests that incorporating these temperature-dependent configurations is necessary to describe structural properties of the material.

Performance summary

MACE-MP-0 correctly captures the loss of Jahn-Teller distortions on heating, with an underestimation of the transition temperature compared with AIMD that is largely corrected with a small amount of fine-tuning.

A.13 Point and extended defects in BCC metals

This test explores bulk and extended defect properties of three prototypical BCC metals: W, Mo and Nb. An accurate description of these properties is essential to enable predictive modelling of mechanical responses to applied loads such as dislocation glide (214), dislocation climb through interaction with point defects, grain boundary motion, and the competition between cleavage and dislocation emission that underpins the brittle to ductile transition in fracture. Across-the-board accuracy for bulk and defect properties in these systems is challenging even for bespoke machine learning potentials fit to carefully curated datasets (215).

	DFT	MACE-MP-0b3	MACE-MP-0b3-FT
W	3.185	3.203	3.191
Mo	3.163	3.182	—
Nb	3.322	3.321	—

Table S1: Lattice constants (in units of Å) for W, Mo and Nb computed by variable-cell minimisation. DFT reference data is from Ref. (216) for W, Ref. (217) for Mo and Ref. (218) for Nb.

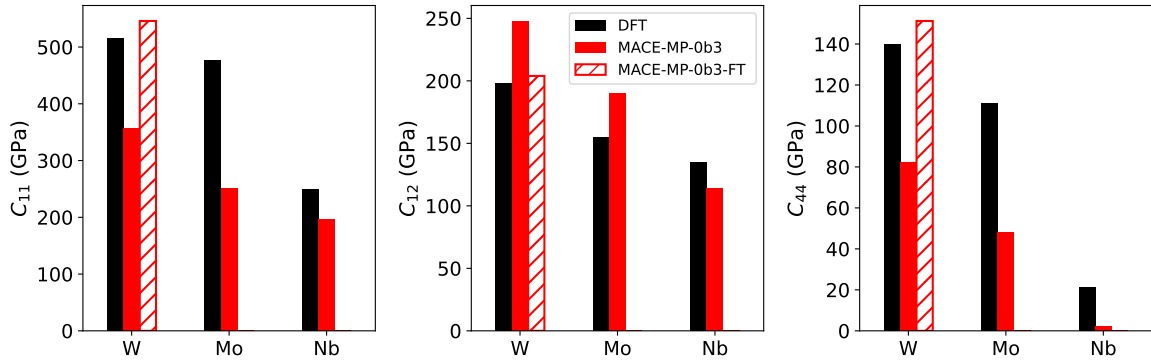


Figure S18: Cubic elastic constants for W, Mo and Nb computed with linear stress-strain fits for DFT (black; from Ref. (216) for W, Ref. (217) for Mo and Ref. (218) for Nb), the MACE-MP-0b3 (solid red). MACE-MP-0b3-FT model finetuned on W data is shown with hatched red bars.

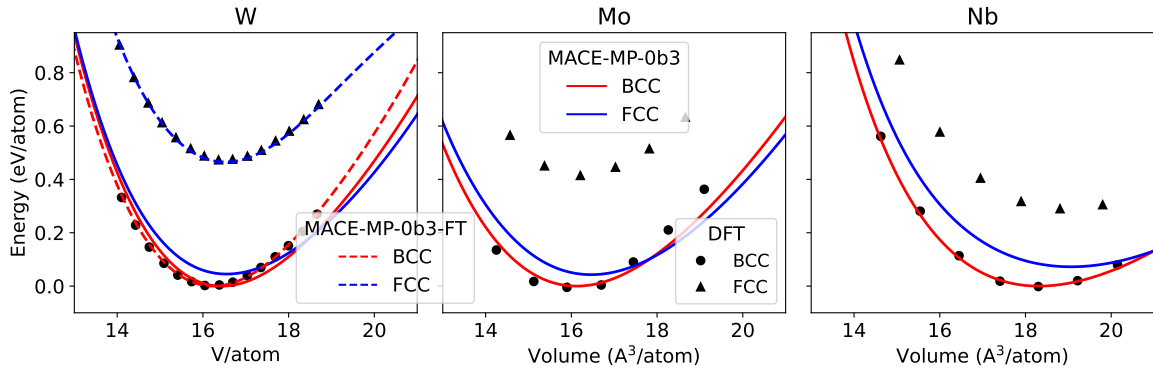


Figure S19: Energy vs volume curves for W, Mo and Nb computed with the medium MACE-MP-0b3 model (solid lines) FCC (red) and BCC (blue) crystals. MACE-MP-0b3-FT model finetuned on W data is shown with dashed lines. The DFT reference data shown with the black circular (BCC) and triangular (FCC) points is from Ref. (219).

Lattice and elastic constants are shown in Table S1 and Fig. S18, respectively. Lattice constants are generally well reproduced, but there is a general softening of the elastic response. In Fig. S19 we compare the energy-volume (E-V) curves predicted by the MACE-MP-0b3 model for BCC and FCC phases of the three metals with reference DFT data from Ref. (219). The BCC cases show generally good agreement. There is room for improvement in the curvature of the E-V curves (critical for the elastic properties) for W and Mo, while Nb is well described. FCC energies are underestimated while the curvature is approximately correct giving reasonable predictions of the elastic response.

We next investigated point defect formation energies, including vacancies and self-interstitial atoms (SIAs). Calculations were performed in a $5 \times 5 \times 5$ supercell and were relaxed to a force tolerance of 1×10^{-5} eV/Å. For the SIAs, a short MD run was performed to escape the initial metastable configuration. The results, illustrated in Fig. S20 show good agreement with reference DFT data is from (220): vacancy energies are predicted within ca. 20% of the DFT value and SIA energies within 50%. For all three elements the MACE-MP-0b3 predicts that the $\langle 111 \rangle$ dumbbell is the most stable SIA configuration, in agreement with DFT.

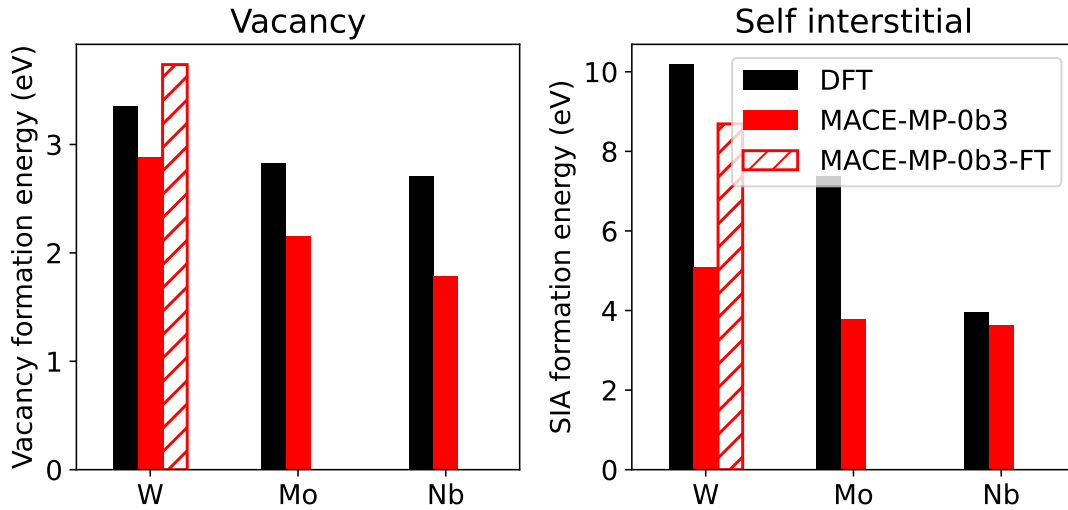


Figure S20: Vacancy and SIA formation energies for W, Mo and Nb computed with the MACE-MP-0b3 model (solid red), W finetuned MACE-MP-0b3-FT model (hatched red) and DFT reference data from Ref. (220) (black).

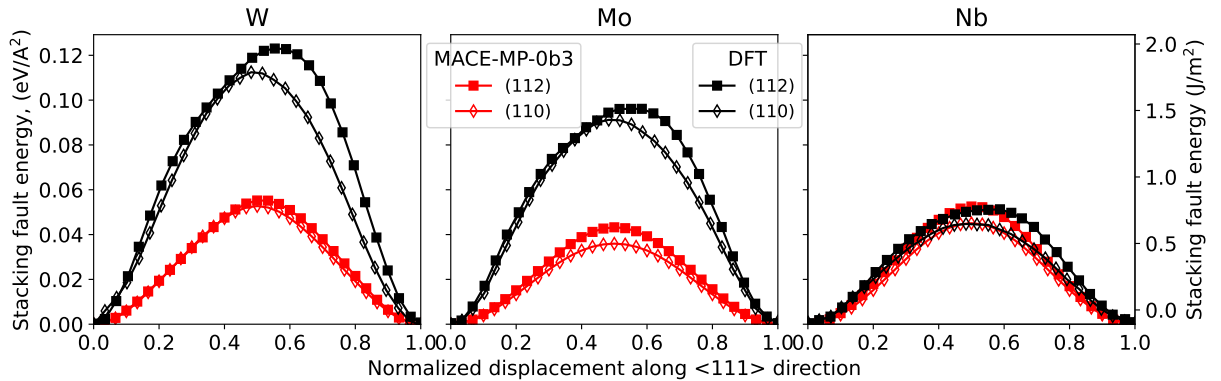


Figure S21: Generalised stacking fault profiles for (112)[111] and (110)[111] Γ -surfaces predicted by the MACE-MP-0b3 model, shown in red. The DFT reference data, shown in black, is from Ref. (221).

We next looked at generalised stacking fault energy profiles for the $\langle 112 \rangle$ and $\langle 110 \rangle$ Γ -surfaces along the $[111]$ direction as shown in Fig. S21. These results were obtained with constrained minimisation where atoms were allowed to move only in the direction perpendicular to the cut surface and with a force tolerance of $1 \times 10^{-3} \text{ eV/\AA}$. The details of the method are explained in Ref. (222). Nb is very well described by the MACE-MP-0b3 model, while there is an underestimate in the stacking fault energies in W and Mo by around a factor of two that explains the underestimates in the screw dislocation glide barriers discussed below.

Dislocations in BCC materials lie predominantly in the $\langle 111 \rangle \{110\}$ and $\langle 100 \rangle \{010\}$ slip systems. We investigate the characteristics of $\langle 111 \rangle$ screw and $\langle 100 \rangle$ edge dislocations by calculating the transition pathways and Peierls barriers using the nudged elastic band (NEB) method and the MACE-MP-0b3 potential comparing against DFT (223) and hybrid QM/MM calculations (224, 225) where the data is available.

The cells contained ≈ 1400 atoms for the $[111]$ screw dislocation and ≈ 2200 for the $[100]$ edge dislocation. Geometry optimisations to obtain starting configurations used the FIRE algorithm with a force tolerance of $1 \times 10^{-6} \text{ eV/\AA}$. To create and analyze atomistic dislocation configurations we employed the `matscipy.dislocation` module (172). The transition path calculation is performed with an adaptive ODE solver (203) following the approach of Refs. (224, 226), using fifteen intermediate images with stopping force tolerance of $0.025 \text{ eV/\AA}$. Starting positions for the NEB relaxation were obtained by linear interpolation between initial and final configurations.

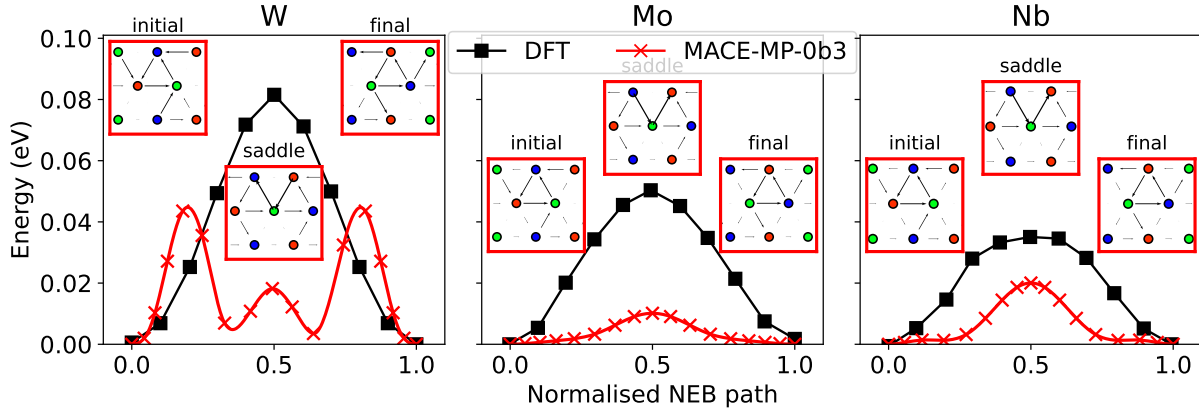


Figure S22: Screw dislocation glide barriers for W, Mo and Nb. DFT data from (223)

Figs. S22 and S23 illustrate the NEB minimum energy path depicting the Peierls barriers for $\langle 111 \rangle$ screw and $\langle 100 \rangle$ edge dislocations in W, Mo and Nb. We compare to DFT results where they are available. Insets within Fig. S22 illustrate the dislocation core structures at the initial, intermediate, and final positions along the MACE-MP-0b3 minimum energy path. Screw dislocations are known to be a sensitive probe of potentials, since the accuracy required is on the meV/atom level. DFT predicts dislocations to move between highly symmetric easy core configuration via split core configuration at the saddle point (223). For all three metals the degenerate core is incorrectly predicted to be the most stable configuration by MACE-MP-0b3 model. At the same time, split core configuration at the saddle point is correctly predicted for Mo and Nb, while for W the single hump shape of the glide barrier is not reproduced with some intermediate configurations having energy, close to the degenerate core. The barrier height is underestimated for all three metals compared to reference DFT results (223). For the edge dislocation, where energy differences are larger, we find that the barrier height aligns well with QM/ML results in W (224) and Mo (225). However, the presence of a minimum along the transition path for Mo results in a spurious stable dislocation configuration.

We anticipate the performance of MACE-MP-0b3 for all properties considered here would be substantially improved by enhanced accuracy in stress and a more precise agreement on the elastic constants, followed by fine-tuning on defect configurations where necessary.